

Math AAHL Paper 2 TZC (Americas)

Worked solutions: <https://www.youtube.com/watch?v=P3ogXGipf2c>

1.

[6]

Leg length (L cm)	85	82	78	91	76	84	80	88
Race time (T seconds)	65.2	67.3	70.0	60.8	72.5	66.1	68.9	62.4

The equation of the regression line T on L for this data can be written in the form $T = aL + b$.

(a) Find the value of a and the value of b . [2]

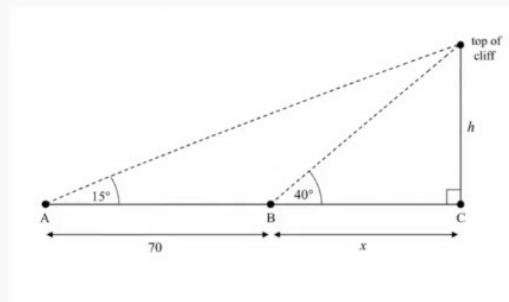
(b) Write down the value of the Pearson's product-moment correlation coefficient, r . [1]

(c) Use the equation of the regression line of t on L to predict the time of a student with leg length 79 cm. [2]

The equation of the regression line of L on T for this data can be written in the form $L = pT + q$.

(d) Find the value of p and the value of q . [1]

2.



(a) Find an expression in terms of h and x for

(i) $\tan 40^\circ$

(ii) $\tan 15^\circ$

(b) Hence, find the value of x .

(c) Find the value of h .

3.

[5]

$V(t) = 5 + 20e^{-kt}$, where $t \geq 0$ and k is a positive constant.

(a) Write down the initial volume inside the balloon.

[1]

After 4 minutes, the volume of air inside the balloon is half the initial volume.

(b) Find the value of k .

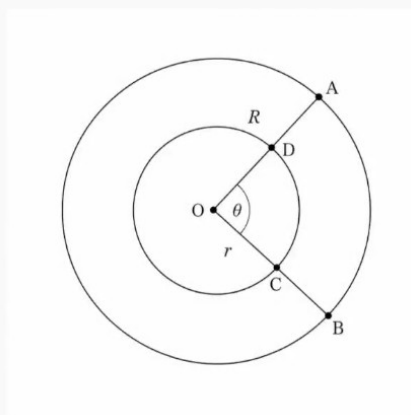
[2]

(c) Find the time at which the volume of air inside the balloon is 8 cubic metres.

[2]

4.

[7]



The area of the sector OAB is 50cm^2 and its perimeter is 40cm . The area of sector ODC is 18cm^2 .

(a) Find the value of θ .

[5]

(b) Hence, find the value of r .

[2]

KINEMATICS

5.

[7]

$$v(t) = (3t^2 - 10t + 4)e^{-0.5t} - 4 \cos t$$

where $0 \leq t \leq 6$.

(a) Find the maximum displacement of the particle.

[4]

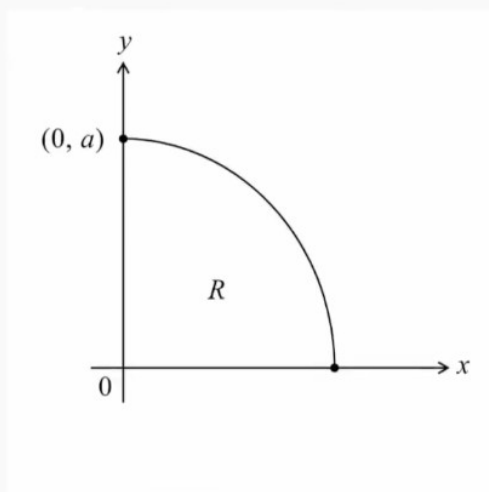
(b) Determine the intervals of time when the acceleration of the particle is negative.

[3]

6.

$$y = \sqrt{9\left(1 - \frac{x^2}{4}\right)}$$

for $0 \leq x \leq 2$.



(a) Write down the value of a .

The region R is bounded by the curve, the x -axis and the y -axis. R is rotated through 360° about the y -axis to form a solid.

(b) Find the volume of this solid.

[5]

RANDOM VARIABLES

7.

[7]

The continuous random variable X has a probability density function f , given by

$$f(x) = \begin{cases} \frac{3}{4}x^2(2-x) & 0 \leq x \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find $P(0.5 \leq X < 1.5)$.

[2]

The discrete random variable Y is defined as the continuous random variable X rounded to the nearest integer. Hence, Y can only take the value 0, 1 or 2. The probability distribution for Y is given in the following table.

y	0	1	2
$P(Y = y)$	a	b	c

(b)

[3]

(i) Write down the value of b .

(ii) Find the value of a and the value of c .

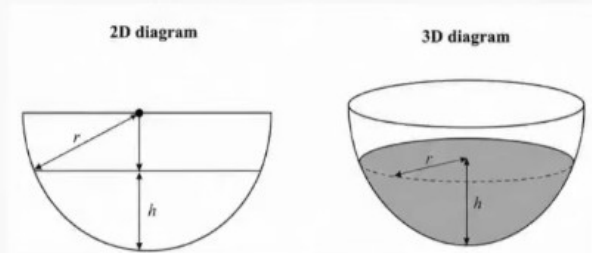
(c) Find $E(Y)$.

RELATED RATES

8.

[7]

~~mn~~ 15 = radius of bowl.



(a) Show that $r = \sqrt{30h - h^2}$.

[2]

Water is being poured into the bowl at a constant rate of 20cm^3 per minute.

The volume of the water in the bowl is given by $V = \frac{\pi}{3} (45h^2 - h^3)$.

(b) Find the rate of change of f when $h = 5\text{cm}$.

COUNTING PRINCIPLES

9.

[6]

Seven girls and three boys sit on a bench in single row for a group photo.

Determine the number of possible arrangements of children, given that

(a) no boy sits next to any other boy

[3]

(b) no boy sits next to any other boy **and** no girl sits at the end of the bench.

[3]

PROBABILITY DISTRIBUTIONS

10.

[18]

The waiting time, T minutes, of patients at a clinic can be modelled by a normal distribution with mean μ and standard deviation σ . The waiting times of all patients are independent of each other.

3.8% of the patients have a waiting time of less than 20 minutes, while 7.3% of the patients have a waiting time of more than 30 minutes.

(a)

[7]

(i) Find the value of μ and the value of σ .

(ii) Given that at least 13 of these patients attended their appointment, find the probability that more than 2 patients did not attend their appointment.

On any day, the doctor can see a maximum of 40 patients.

Some patients do not attend their appointments. The clinic decides to allow more than 40 appointments to be made each day.

The clinic aims for a probability of at least 99% that the doctor sees all the patients who arrive for their appointment.

(c) Find the greatest number of appointments that can be made to achieve the clinic's aim. [4]

RATIONAL FUNCTIONS

11.

[17]

The function f is defined by

$$f(x) = \frac{x^2 + bx + c}{x + d},$$

where $x \in \mathbb{R}$, $x \neq -d$ and $b, c, d \in \mathbb{Z}$.

The graph of $y = f(x)$ has asymptotes $x = 3$ and $y = x + 7$.

(a) Write down the value of d .

[1]

The graph of $y = f(x)$ has asymptotes $x = 3$ and $y = x + 7$.

(b) Show that $b = 4$.

The graph of $y = f(x)$ has a local maximum at $x = -3$.

(c) Find the value of c .

[4]

(d) Sketch the graph of $y = f(x)$ for $-30 \leq x \leq 30$. Show the values of any axis intercepts and the coordinates of any local maxima and minima, and label the asymptotes. [4]

The equation $|f(x)| = k$ has four solutions.

(e) Find the range of possible values of k .

[2]

The graph of $y = g(x)$ is obtained from the graph of $y = f(x)$ by a translation of $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

(f) Find the equations of the vertical asymptotes of $y = \frac{1}{g(x)}$. [4]

DIFFERENTIAL EQUATIONS

12. [18]

Consider the differential equation

$$\frac{dy}{dx} = y^3 - \frac{y}{x},$$

for $x, y > 0$. It is given that $y = 2$ when $x = 1$.

(a) Use Euler's method with a step length of 0.01 to find an approximate value of y when $x = 1.05$. [3]

By using the substitution $\frac{1}{y^2} = -2z$, show that the differential equation can be expressed as $\frac{dz}{dx} - \frac{2z}{x} = 1$.

(b) Show that the differential equation can be expressed as $\frac{dz}{dx} - \frac{2z}{x} = 1$. [4]

(c) Hence, solve the differential equation, giving your answer in the form $y = f(x)$. [7]

(d) Find the domain and range of $f(x)$. [3]

(e) By considering the shape of the graph of $f(x)$, explain whether Euler's method gives an overestimate or underestimate of the value of y when $x = 1.05$. [1]